Analyser connection checklist

Information we need to connect your lab machines to Health Care Labs

Fill one sheet per analyser
Photos of the settings screens are welcome

How it works, in short. A small program (the "connector") runs on one Windows PC in your lab. Each analyser sends its results to that PC over a network cable or a serial cable. The connector forwards them over the internet to Health Care Labs, where they appear in result entry next to the matching bill (found by the tube barcode).

1 About the analyser

Make and model For example Mindray BS-240, Sysmex XN-330, Erba EM-200	
Department Biochemistry, haematology, immunoassay, urine, coagulation...	
Software version Usually on the About or System screen	
Does it read tube barcodes?	Yes / No
Is it connected to any lab software today? If yes, which one (for example Flabs)	

2 How it is connected

On the analyser, open the **LIS**, **Host** or **Communication** settings screen. Please send a **photo of that screen** and fill in the part that applies.

Type of cable to the lab PC	Network (LAN) cable / Serial cable (9-pin RS-232 or USB-serial) / Exports result files / Not sure
Protocol The message format	ASTM / HL7 / Other: _____ / Not sure
Is the LIS / host communication switched on?	Yes / No

If a network (LAN) cable:

Analyser's IP address	
Host / LIS IP address set on the analyser	
Port number	
Does the analyser connect to the host, or wait for the host to connect? Often shown as "Client" or "Server" mode	Analyser connects (client) / Analyser waits (server) / Not sure

If a serial cable:

Baud rate (speed)	9600 / 19200 / 38400 / 115200 / other: ____
Data bits, parity, stop bits	for example 8, None, 1
Cable type 9-pin to 9-pin, or a USB-serial adapter on the PC	

If it exports result files: the folder they are saved to, and one sample file.

3 Test codes

The short code the analyser uses for each test, so we can match it to your test list. A printout or photo of the analyser's test list is fine.

Test name in the lab	Code on the analyser
For example: Glucose fasting	For example: GLU

4 The lab PC that will run the connector

Which PC is cabled to, or on the same network as, this analyser?	
Windows version	Windows 10 / Windows 11 / other: ____
Does it stay on during all lab working hours?	Yes / No
Does it have internet?	Yes / No
PC's IP address on the lab network Needed only when the analyser connects to the PC. Ideally fixed, so it does not change	
Can someone sign in as administrator on it? Needed once, for the install	Yes / No
Is the analyser vendor's own software running on it? It may already be using the same COM port	Yes / No

5 Contact for the setup day

Lab contact who can work the analyser	
Analyser service engineer (vendor), if any	
Person who manages the lab's computers and network	

6 Installing the connector (on setup day)

1. **Add the analyser in the app.** Quality Control > QC Setup > Machines > Add machine. In Tests, set each test's **machine code** to the codes from section 3.
2. **Install Node.js** on the lab PC from `nodejs.org`: the "LTS" version, default options. One time only.
3. **Download the connector** from the app (Connect analyser window) or `laboratory-management-systems.web.app/downloads/hcl-connector.zip`. Right-click > Extract All, to `C:\HCL-Connector`.
4. **Get the settings.** In the app, open the machine > **Connect analyser** > Make a token > choose how it is linked and fill in the port details > **Copy**.
5. **Run setup.** In `C:\HCL-Connector`, double-click **setup.bat**. Notepad opens: paste the settings over everything, save, close. Setup checks the token, asks for administrator permission, and installs the connector so it starts with Windows.
6. **Check.** The status page opens at `http://localhost:7300`; the analyser should show **listening** or **connected**. In the app, the machine shows **Connected** within a minute.
7. **Test one sample.** Make a bill, put its barcode on a tube, run it on the analyser. The values appear in result entry under **Machine value**.

On the analyser (network, analyser connects): set the host IP to the lab PC's IP and the same port as in the app. **On the analyser** (serial): keep its speed and settings exactly as entered in the app.

Results are matched by the tube barcode on the bill. If the internet drops, results wait on the lab PC and are sent automatically when it returns. Anything that cannot be matched is kept in the app for checking, never lost.